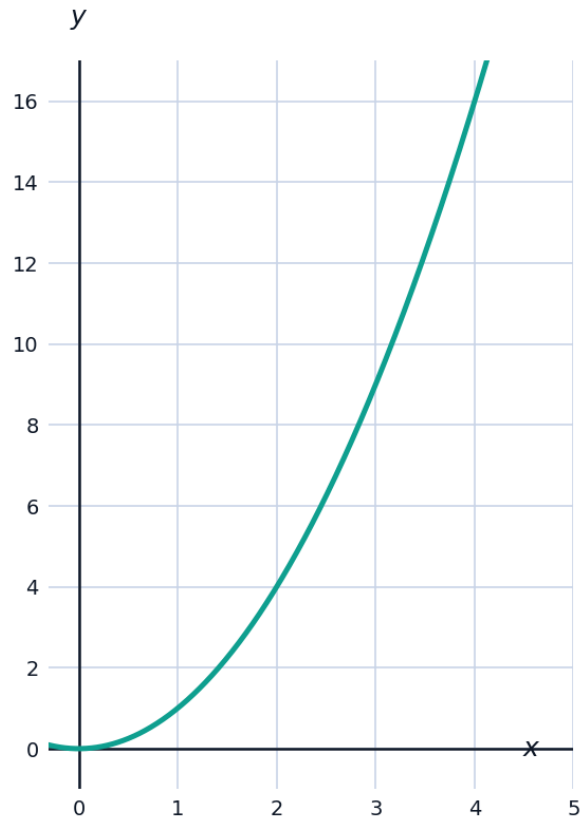


Riemann Sums and Definite Integrals

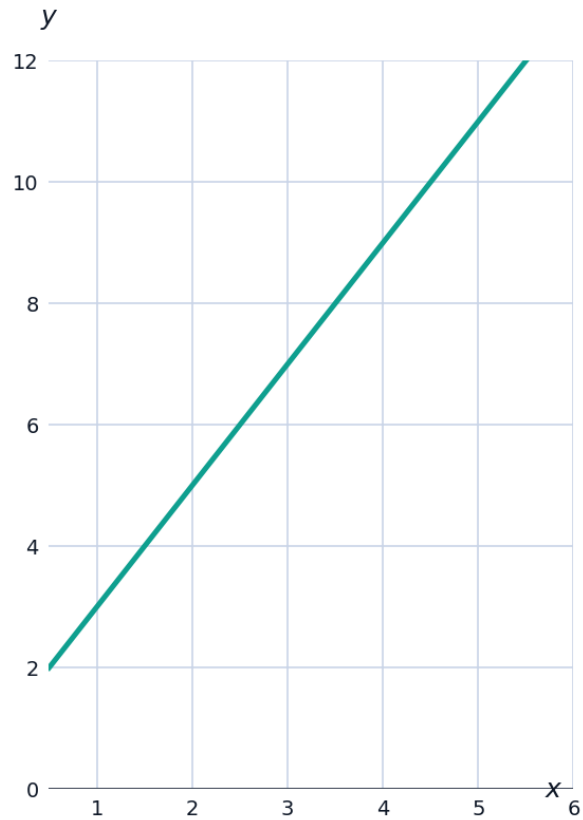


Approximating area with Riemann sums

- 1 Estimate $\int_0^4 x^2 dx$ using a right-endpoint Riemann sum with $n = 4$ subintervals.



- 2 Estimate $\int_1^5 (2x + 1) dx$ using a left-endpoint Riemann sum with $n = 4$ subintervals.



The definite integral and its properties

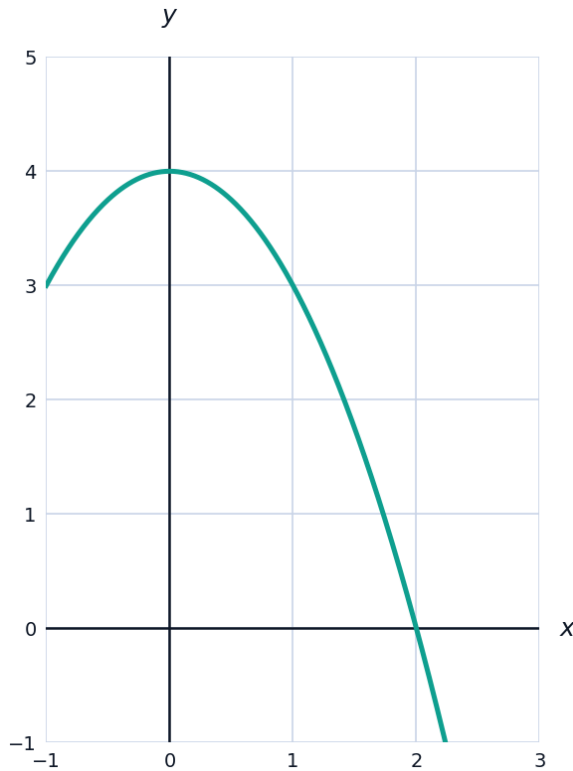
- 3 Given $\int_0^3 f(x) dx = 8$ and $\int_3^7 f(x) dx = -2$, find:

a $\int_0^7 f(x) dx$

b $\int_7^3 f(x) dx$

c $\int_0^3 5f(x) dx$

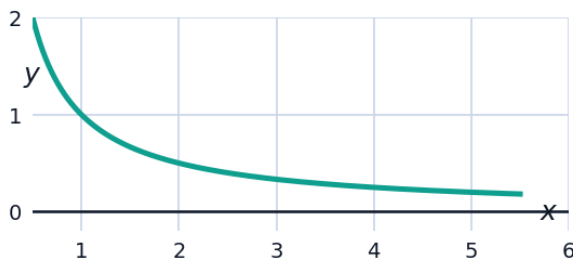
- 4 The graph of $f(x) = 4 - x^2$ is shown, together with the region between the curve and the x -axis on $[0, 2]$. Evaluate $\int_0^2 (4 - x^2) dx$ exactly.



- 5 Evaluate $\int_{-2}^2 (x^3 - 4x) dx$ by recognizing the integrand's symmetry, without computing an antiderivative.

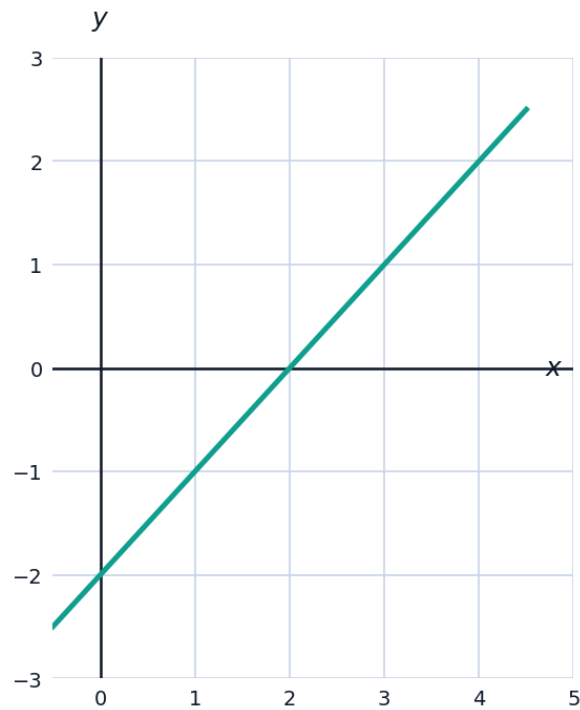
Midpoint and trapezoidal approximations

- 6 Estimate $\int_0^4 x^2 dx$ using a midpoint Riemann sum with $n = 4$ subintervals, and compare to the exact value $\frac{64}{3} \approx 21.33$.
- 7 Estimate $\int_1^{5.1} \frac{1}{x} dx$ using the trapezoidal rule with $n = 4$ subintervals, to three decimal places.

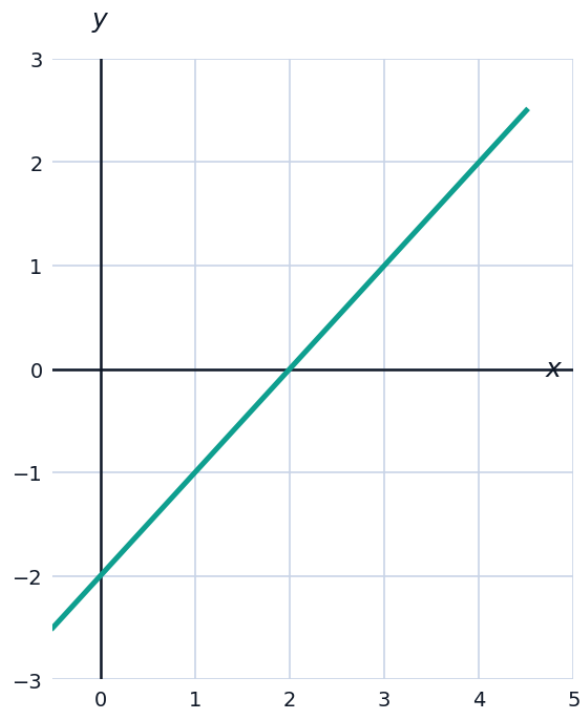


Signed area

- 8 The graph of $f(x) = x - 2$ crosses the x -axis at $x = 2$. Evaluate $\int_0^4 (x - 2) dx$ and explain why the result does not equal the total area between the curve and the axis.



- 9 Find the total (unsigned) area between $f(x) = x - 2$ and the x -axis on $[0, 4]$, splitting the integral at the sign change $x = 2$.



Comparing Riemann sum estimates

- 10 For the increasing function $f(x) = x^2$ on $[0, 4]$, will the left-endpoint sum overestimate or underestimate the true area? Justify your answer.
- 11 Using the right-endpoint result (30) and left-endpoint result (24, computed earlier for a different function) is not directly comparable — instead, compare the right-endpoint estimate (30) for $\int_0^4 x^2 dx$ to the exact value $\frac{64}{3} \approx 21.33$, and explain the direction of the error.
-

Riemann sums from a table of values

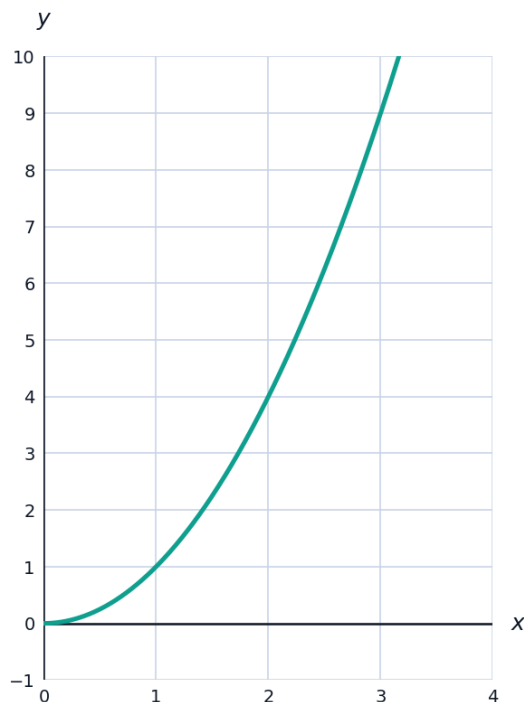
- 12 A table gives $f(0) = 2$, $f(1) = 5$, $f(2) = 9$, $f(3) = 14$, $f(4) = 20$. Use a right-endpoint Riemann sum with these 4 equal subintervals of width 1 to approximate $\int_0^4 f(x) dx$.
-

The definite integral as a limit of Riemann sums

- 13 Write the formal limit definition of the definite integral $\int_a^b f(x) dx$ in terms of a Riemann sum, and explain the role of $n \rightarrow \infty$.
-

Riemann sums with sigma notation

- 14 Write the right-endpoint Riemann sum R_n for $\int_1^3 x^2 dx$ using sigma notation, with $\Delta x = \frac{2}{n}$ and $x_i = 1 + i\Delta x$. Do not evaluate the sum.



- 15 Estimate $\int_0^6 f(x) dx$ using a midpoint Riemann sum with $n = 3$ subintervals, given $f(1) = 4$, $f(3) = 10$, $f(5) = 16$.
-

Comparing approximation methods on a concave function

- 16 For a concave-down function on $[a, b]$, which is larger: the trapezoidal estimate or the midpoint estimate? Justify briefly, and state which one overestimates the true area.

